

Practice with multiple continuous variables in R, Scatter Plots and Correlation

1 Multiple Continuous Variables in R

In this notes, we will practice with multiple continuous variables in R, focusing on scatter plots and correlation. We will start by loading the necessary libraries and the “airquality” dataset.

```
library(tidyverse)
library(krulRutils)
library(GGally)
library(here)
options(scipen = 999) # Disable scientific notation
data(airquality)
```

We start by doing a simple cleaning of our dataset, removing rows with missing values and collecting the month and day into a single date column:

```
airquality_tbl <- airquality %>%
  as_tibble() %>%
  drop_na() %>%
  mutate(Date = as.Date(paste(1973, Month, Day, sep = "-"))) %>%
  select(-Month, -Day) %>%
  glimpse()
```

Rows: 111

Columns: 5

```
$ Ozone    <int> 41, 36, 12, 18, 23, 19, 8, 16, 11, 14, 18, 14, 34, 6, 30, 11, ~
$ Solar.R  <int> 190, 118, 149, 313, 299, 99, 19, 256, 290, 274, 65, 334, 307, ~
$ Wind     <dbl> 7.4, 8.0, 12.6, 11.5, 8.6, 13.8, 20.1, 9.7, 9.2, 10.9, 13.2, 1~
$ Temp     <int> 67, 72, 74, 62, 65, 59, 61, 69, 66, 68, 58, 64, 66, 57, 68, 62~
$ Date     <date> 1973-05-01, 1973-05-02, 1973-05-03, 1973-05-04, 1973-05-07, 1~
```

We now construct a matrix of scatter plots to visualize the relationships between the continuous variables in our dataset. The `ggpairs` function from the `GGally` package is particularly useful for this purpose:

```
ggpairs_plot <- (airquality_tbl %>%  
  select(-Date) %>% # Exclude the Date column for scatter plots  
  ggpairs() +  
  labs(  
    title = "Scatter Plot Matrix of Air Quality Data",  
    subtitle = "Showing relationships between Ozone, Solar.R, Wind, and Temp"  
  )  
) %>%  
  theme_krul_ggpairs()  
  
ggsave(  
  filename = here("images", "ggpairs_airquality.png"),  
  plot = ggpairs_plot,  
  width = 12,  
  height = 12,  
  dpi = 300  
)  
  
ggpairs_plot
```

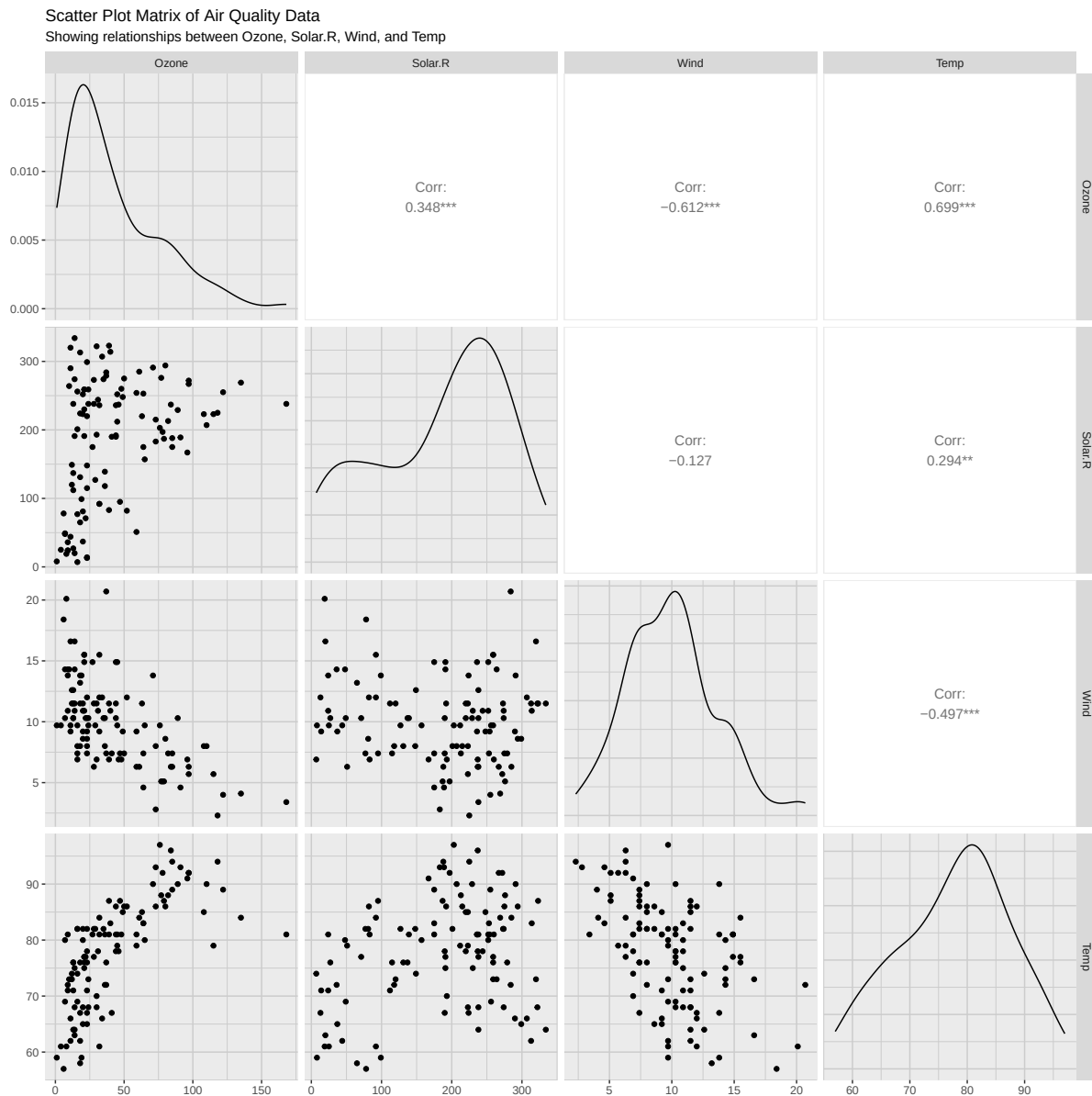


Figure 1: Scatter plot matrix of the airquality dataset

We notice that the variables with the strongest correlation are “Ozone” and “Temperature”. So we now focus on these two variables to create a more detailed scatter plot with a fitted regression line:

```
ozone_temp_plot <- airquality_tbl %>%
  ggplot(aes(x = Temp, y = Ozone)) +
  geom_smooth()
```

```

    method = "lm",
    color = c_pal("C blue"),
    se = FALSE,
    linewidth = 0.5
) +
geom_point(
  alpha = 1,
  color = c_pal("C red"),
  size = 1
) +
labs(
  title = "Scatter Plot of Ozone vs Temperature",
  x = "Temperature (°F)",
  y = "Ozone (ppb)"
) +
theme_krul()

ozone_temp_plot

```

`geom\_smooth()` using formula = 'y ~ x'

Scatter Plot of Ozone vs Temperature

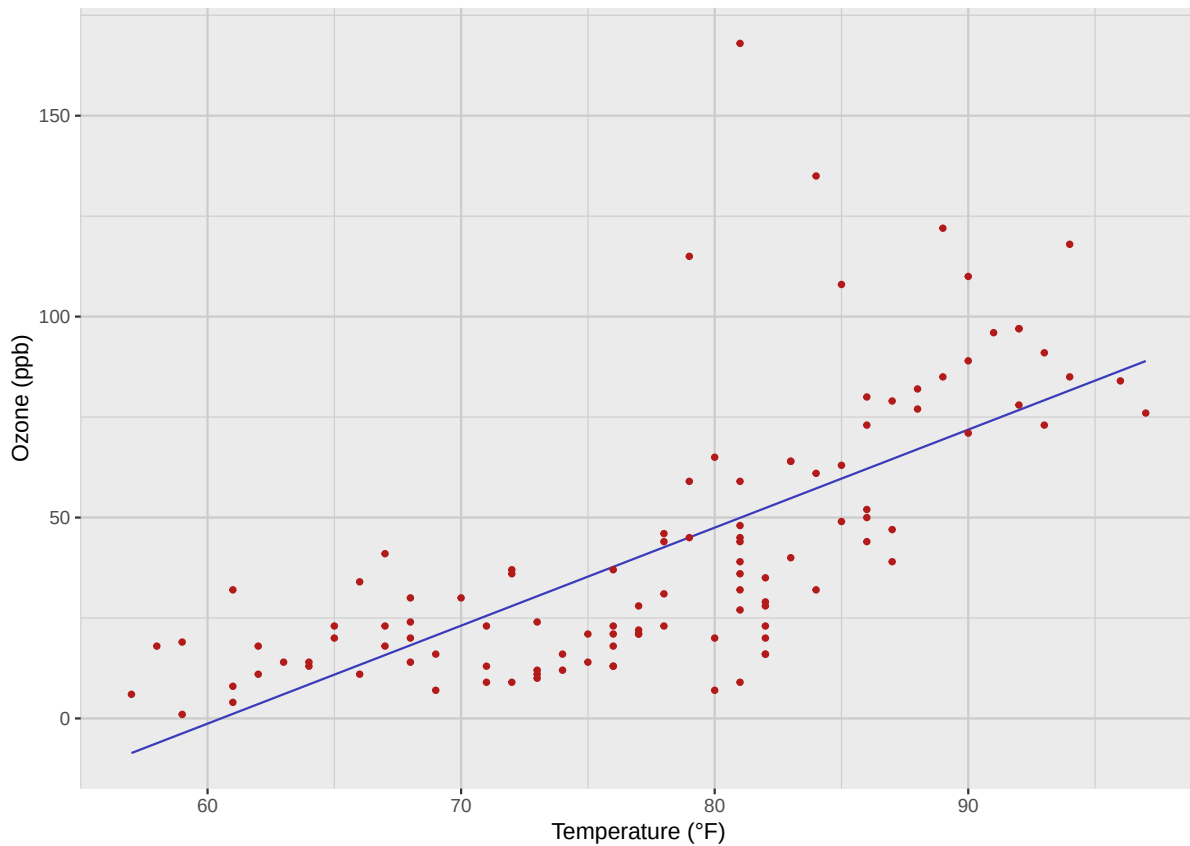


Figure 2: Scatter plot of Ozone vs Temperature with regression line

Finally, we show a time series showing the temperature in New York City during the months of May to September 1973, which is the period covered by the “airquality” dataset:

```
temp_time_series_plot <- airquality_tbl %>%  
  ggplot(aes(x = Date, y = Temp)) +  
  geom_line(color = c_pal("C blue"), linewidth = 0.5) +  
  labs(  
    title = "Temperature in New York City (1973)",  
    x = "Date",  
    y = "Temperature (°F)"  
  ) +  
  theme_krul()  
  
temp_time_series_plot
```

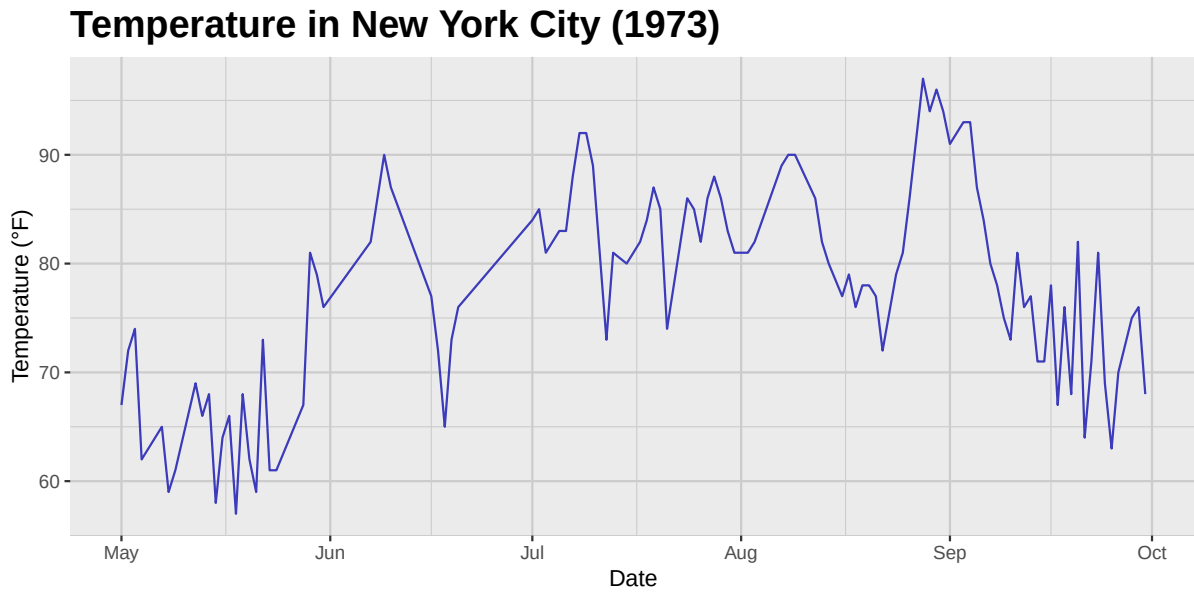


Figure 3: Temperature in New York City from May to September 1973